

NORTH RONA ISLAND, United Kingdom

WMO: 030110

Lat: 59.121N

Lon: 5.815W

Elev: 103

StdP: 100.09

Time Zone: 0.00 (EUW)

Period: 94-09

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	0.1	1.1	-6.8	2.1	2.0	-5.6	2.4	2.4	32.1	5.7	29.0	6.7	12.9	350	1.137

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	2.0	16.0	14.2	14.7	13.3	14.1	12.8	14.5	15.4	13.7	14.4	13.1	13.7	6.5	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.0	10.1	15.0	13.2	9.6	14.2	12.6	9.2	13.6	40.9	15.6	38.7	14.5	37.1	13.8	21.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.6	24.7	22.2	DB	-1.7	18.8	1.4	3.1	-2.8	21.1	-3.6	22.9	-4.4	24.7	-5.4	26.9	(4)
(5)			WB	-3.1	16.4	1.1	2.0	-3.9	17.8	-4.5	19.0	-5.1	20.1	-5.8	21.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.3	5.7	5.2	5.3	6.6	8.1	10.0	11.9	12.2	11.1	9.6	7.6	6.2
	DBStd	3.12	1.73	2.13	1.98	1.97	1.88	1.52	1.63	1.61	1.88	1.95	2.09	2.30
	HDD10.0	832	133	134	146	102	64	19	2	2	11	30	73	117
	HDD18.3	3654	391	367	404	351	318	251	201	189	217	269	322	375
	CDD10.0	221	0	0	0	1	5	18	59	72	44	19	2	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	11.1	14.1	13.8	12.9	10.9	8.8	8.9	7.9	7.7	10.0	12.3	12.9	13.1
Precipitation	PrecAvg	1797	218	174	165	103	90	91	101	116	150	185	194	209
	PrecMax	2490	395	365	403	162	211	147	151	234	238	284	322	363
	PrecMin	1262	82	70	54	52	17	44	46	54	50	82	91	55
	PrecStd	256	82	84	67	28	38	28	24	42	47	54	57	79
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.2	9.7	9.6	13.7	14.8	15.3	17.9	18.7	15.4	14.3	12.8	11.2
		MCWB	8.9	8.6	8.4	10.1	11.2	13.1	16.1	16.1	13.4	12.5	11.6	9.9
	2%	DB	9.5	9.2	8.8	10.0	12.3	13.5	15.9	16.2	14.3	13.3	11.4	10.5
		MCWB	8.4	8.3	7.7	8.6	10.4	12.0	14.5	14.7	13.0	11.9	10.3	9.4
	5%	DB	9.0	8.8	8.4	9.3	11.0	12.5	14.7	14.7	13.7	12.6	10.8	10.0
		MCWB	7.9	7.8	7.3	8.1	9.6	11.2	13.4	13.5	12.6	11.4	9.7	9.0
	10%	DB	8.4	8.3	8.0	8.8	10.2	11.8	13.9	14.0	13.2	12.1	10.4	9.4
		MCWB	7.2	7.2	6.8	7.7	8.9	10.6	12.8	12.9	12.2	10.9	9.2	8.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.6	9.1	8.7	10.2	11.6	13.6	16.1	15.9	14.1	12.9	11.9	10.5
		MCDB	10.1	9.4	9.2	12.4	13.7	15.1	17.7	17.6	14.8	13.6	12.5	10.8
	2%	WB	8.7	8.6	8.1	9.0	10.5	12.1	14.7	14.5	13.4	12.3	10.7	9.7
		MCDB	9.2	9.0	8.7	9.7	12.0	13.1	15.7	15.5	14.0	13.0	11.3	10.2
	5%	WB	8.1	8.1	7.6	8.4	9.7	11.5	13.7	13.7	12.9	11.7	10.0	9.2
		MCDB	8.8	8.6	8.3	9.2	10.9	12.3	14.5	14.3	13.5	12.5	10.6	9.9
	10%	WB	7.4	7.4	7.0	7.9	9.1	10.9	12.9	13.0	12.4	11.1	9.4	8.5
		MCDB	8.3	8.1	7.9	8.7	10.1	11.6	13.6	13.7	13.2	12.0	10.2	9.4
Mean Daily Temperature Range	5% DB	MDBR	2.9	2.8	2.7	2.4	2.3	2.0	2.0	2.0	2.2	2.3	2.6	2.8
		MCDBR	3.6	3.1	3.0	2.7	3.3	3.0	3.3	2.9	2.7	2.6	2.7	3.0
		MCWBR	4.1	3.5	3.2	2.4	2.6	2.7	2.8	2.7	2.6	2.8	3.3	3.7
	5% WB	MCDBR	3.6	3.0	3.2	2.6	3.2	2.8	3.3	2.8	2.4	2.7	2.8	3.1
		MCWBR	4.2	3.5	3.5	2.4	2.7	2.7	2.8	2.7	2.6	3.1	3.5	3.9
Clear-Sky Solar Irradiance	taub	0.337	0.351	0.364	0.379	0.373	0.367	0.373	0.364	0.361	0.346	0.322	0.308	
	taud	2.109	2.235	2.310	2.326	2.405	2.464	2.461	2.492	2.459	2.349	2.182	2.070	
	Ebn at Noon	426	645	766	823	857	869	853	836	772	651	448	322	
	Edh at Noon	52	74	92	106	105	101	99	89	79	65	46	37	
All-Sky Solar Radiation	RadAvg	0.28	0.85	1.98	3.53	4.79	4.83	4.43	3.67	2.36	1.17	0.41	0.18	
	RadStd	0.03	0.09	0.13	0.27	0.38	0.40	0.36	0.25	0.15	0.10	0.04	0.02	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (13 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page